



JS Built-in Objects

1. Date
2. String
3. JS Global functions

- Basics
 - Used for dealing with date and time
 - Stores number of milliseconds from 1 Jan 1970
- Creating date object
 - Various ways by which date can be created
 - I. no-arg constructor: creates date with current date and time
 - II. now() function: gives no of milliseconds for current date and time
 - III. constructor with int argument: Argument representing no of milliseconds converted to specific date and time
 - IV. constructor with string argument: Date in string format(only few accepted formats) converted to date object
 - V. constructor with date parts: accepts minimum 2 and maximum 7 arguments as different parts starting from year to milisecinds

- Date functionality
 - Getter methods
 - Used for extracting each individual part of the date
 - Setter methods
 - Used for setting new value to each individual part of the date

- Formatting dates
 - Date can be formatted into string using by using
 - i. `toString()`
 - ii. `getTimeString()`
 - iii. `toLocaleString()`

- Comparing dates
 - Using comparison operators like `<`, `>`, `<=`, `>=`.
 - Convert both the dates to milliseconds using `getTime()` function
- Date formats
 - ISO Date: YYYY-MM-DDTHH:MM:SSZ
 - Javascript short dates: MM/DD/YYYY
 - Javascript long dates: MMM DD YYYY
- Time zone
 - Javascript considers time zone as that of browser's time zone by default

- Represents sequence of characters including alphabets, digits, special characters
- Creating string
 - String can be created by wrapping text within
 - single quotes
 - double quotes
 - backticks: Allows to include expressions or variables in the string
- String functionality
 - Getting character at particular index
 - i. `str.charAt(index)`
 - Getting ascii value at particular index
 - i. `str.charCodeAt(index)`

- Changing the cases
 - i. `str.toLowerCase()`
 - ii. `str.toUpperCase()`
- Searching within a string
 - i. `str.indexOf("string to be searched")`
 - ii. `str.lastIndexOf("string to be searched")`
 - iii. `str.search(regex)`
- Getting a section of the string
 - i. `slice(startindex, lastindex)`
 - ii. `substring(startindex, lastindex)`
 - iii. `substr(startindex, length)`

- Concatenating a string
 - i. `str.concat(str1)`
- Splitting string
 - i. `str.split("string at which to split")`
- Replacing string
 - i. `str.replace("to be searched","to be replaced")`

- Global functions can be called without any explicit object or can be called with window object
 - i. `isNaN()`
 - ii. `isFinite()`
 - iii. `parseInt()`
 - iv. `parseFloat()`
 - v. `encodeURIComponent()`
 - vi. `decodeURI()`
 - vii. `eval()`